UNIT - V

Moment Distribution Method

1. Introduction

- The **moment distribution method (MDM)** was developed by **Prof. Hardy Cross** in **1930** (ASCE Transactions, 1932). It is an **iterative displacement method** — a convenient hand-calculation form of the slope-deflection method.
- It avoids solving simultaneous equations: the end moments are obtained by a series of **successive approximations** (cycles) that converge quickly.
- Used for **continuous beams and rigid frames** ; popular in design offices before computers; still used for quick checks.

Basic idea

Lock all joints against rotation → calculate fixed end moments → release one joint at a time, **distribute the unbalanced moment** to the members meeting at the joint, **carry over** half to the far ends → repeat until the unbalanced moments become negligible.

Assumptions

- Prismatic members (constant EI), linear elastic behaviour, small deflections.
- Axial and shear deformations are neglected; joints are rigid.
- Applicable directly to structures **without sway** ; sway frames need an extra correction.

2. Basic Definitions

Term	Meaning
Fixed end moment (FEM)	moment developed at the ends of a loaded member when both its ends are fixed (Unit III table)
Unbalanced moment	algebraic sum of the FEMs (or current end moments) of the members meeting at a joint
Stiffness (K)	moment required to produce a **unit rotation** at one end, the far end being fixed ($4EI/L$) or hinged ($3EI/L$)
Relative stiffness	ratio of stiffnesses; I/L can be used when E is constant
Distribution factor (DF)	fraction of the unbalanced moment taken by a member: $DF = K_{member} / \Sigma K$ at the joint
Distributed (balancing) moment	− (unbalanced moment) × DF

Term	Meaning
Carry-over factor (COF)	ratio of the moment induced at the far fixed end to the moment applied at the near end = ½
Carry-over moment	COF × distributed moment, placed at the far end

Stiffness and carry-over (from slope-deflection)

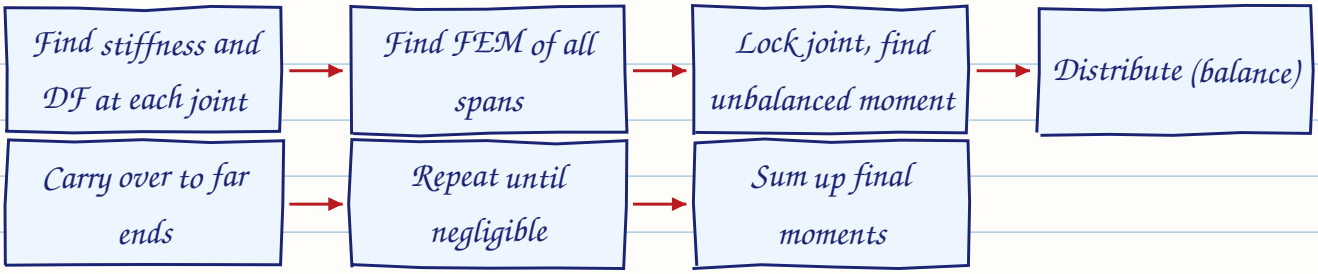
Far end	Stiffness K	Carry-over factor
Fixed	$4EI / L$	+ 1/2
Hinged / roller (simply supported)	$3EI / L$ (= ¾ × $4EI/L$)	0
Free end (cantilever)	0	—
Symmetric loading, symmetric span	$2EI / L$	—
Antisymmetric loading, symmetric span	$6EI / L$	—

- **Distribution factors at a joint always add up to 1.0** (check!).
- **DF of a fixed support = 0** (infinite stiffness — it does not rotate); **DF of a hinged/roller end support = 1.0** .
- Overhanging cantilever: moment at the support is **known** ; it enters as an unbalanced moment and has no DF.

3. Sign Convention

- **Clockwise end moments (acting on the member ends)** are **positive** ; anticlockwise negative (same as slope deflection).
- Moment acting **on the joint** from a member is opposite in sense to the moment acting on the member end.
- At a joint the sum of the member end moments must be **zero** at the end of the process (equilibrium).
- Balancing moment has the **opposite sign** of the unbalanced moment.
- For the final BMD: end moments of a member in the usual convention — **hogging** at an interior support for gravity loads.

4. Procedure of Moment Distribution



1. **Stiffness and DF:** compute K for every member at each joint ($3EI/L$ if far end hinged; $4EI/L$ otherwise); find $DF = K/\Sigma K$.
2. **FEM:** for all loaded spans (clockwise +ve). For a settlement, add $FEM = \pm 6EI\Delta/L^2$.
3. Set up a **table** with a column for each member end (AB, BA, BC, CB ...).
4. **Balance** each joint in turn: unbalanced moment = Σ moments at the joint; balancing moment = − unbalanced × DF; write under each member and draw a line.
5. **Carry over** half of each balancing moment to the other end of the member (not for hinged far end).
6. Continue until the **carry-over is negligible** (say less than 1% of the FEM).
7. **Add each column** to get final end moments; check ΣM at joints = 0.
8. Compute **reactions** (R = free reaction $\pm (M_{left} - M_{right})/L$), SFD and BMD.

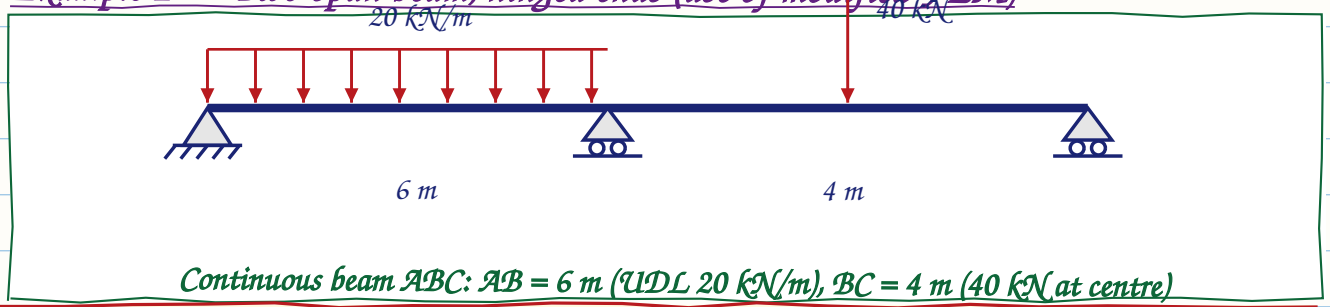
→ Shortcut: when the far end is **hinged**, use the **modified FEM** $F' = F - \frac{1}{2}F_{far}$ and stiffness $3EI/L$, and **do not carry over** to that end. A fixed end is never balanced ($DF = 0$); an **end hinged support** can be balanced once at the start.

Order of balancing

- Any order is allowed, but balance the joint with the **largest unbalanced moment** first; it converges faster. For a joint where the unbalanced moment is zero skip it.

5. Application to Continuous Beams — Examples

Example 1 — Two-span beam, hinged ends (use of modified FEM)



Continuous beam ABC: AB = 6 m (UDL 20 kN/m), BC = 4 m (40 kN at centre)

Problem

Same beam as Unit IV (EI constant). A and C simply supported.

- Both far ends hinged: $K_{BA} = 3EI/6 = 0.5\ EI$; $K_{BC} = 3EI/4 = 0.75\ EI$. **$DF_{BA} = 0.5/1.25 = 0.4$; $DF_{BC} = 0.6$.**
- FEM: AB = − 60, BA = + 60, BC = − 20, CB = + 20. Modified: $F'_{BA} = 60 + 30 =$ **+ 90** , F'_{BC}

= − 20 − 10 = **− 30** (no carry-over to A or C).

Joint / member	BA	BC
DF	0.4	0.6
Modified FEM	+ 90	− 30
Unbalanced at B	+ 60	
Balancing moment	− 24	− 36
Final moment	**+ 66**	**− 66**

- **M_B = 66 kN·m (hogging)** √ same as the slope-deflection and three-moment results. Reactions: R_A = 49 kN, R_B = 107.5 kN, R_C = 3.5 kN.

Example 2 — Beam with a fixed end (single cycle)

Problem

ABC; A fixed, B and C rollers; AB = BC = 6 m; UDL 20 kN/m on both spans.

- $K_{BA} = 4EI/6 = 0.667\,EI$, $K_{BC} = 3EI/6 = 0.5\,EI$ (C hinged). **DF_{BA} = 0.571, DF_{BC} = 0.429.**
- FEM: AB = − 60, BA = + 60, BC (modified) = − 60 − 30 = − 90.

Member end	AB	BA	BC
DF	—	0.571	0.429
FEM	− 60	+ 60	− 90
Balance B (unbalanced − 30)		+ 17.14	+ 12.86
Carry-over	+ 8.57		
Final	**− 51.43**	**+ 77.14**	**− 77.14**

- √ Equal to the slope-deflection result (Unit IV, Example 2).

Example 3 — Three spans, two joints to balance (several cycles)

Problem

Beam ABCD; A fixed; B, C and D rollers; AB = BC = CD = 6 m; UDL 20 kN/m on all spans.

- D is hinged → modified stiffness for CD (3EI/6) and modified FEM F'_{CD} = − 60 − 30 = − 90. $K_{BA} = K_{BC} = 4EI/6$ → **DF_B = 0.5, 0.5** ; $K_{CB} = 4EI/6$, $K_{CD} = 3EI/6$ → **DF_C = 0.571 (CB), 0.429 (CD)** .
- Joint B is balanced at the start (+ 60 − 60 = 0), so begin with C.

Step	AB	BA	BC	CB	CD
DF	—	0.500	0.500	0.571	0.429
FEM	-60.00	+60.00	-60.00	+60.00	-90.00
Bal C(1)				+17.14	+12.86
CO(1)			+8.57		

Step	AB	BA	BC	CB	CD
Bal B(2)		-4.29	-4.29		
CO(2)	-2.14			-2.14	
Bal C(2)				+1.22	+0.92
CO(2c)			+0.61		
Bal B(3)		-0.31	-0.31		
CO(3)	-0.15			-0.15	
Bal C(3)				+0.09	+0.07
CO(3c)			+0.04		
Final (≈)	-62.30	+55.41	-55.36	+76.16	-76.16
Exact	-62.31	+55.38	-55.38	+76.15	-76.15

- **Final:** $M_{AB} = -62.31$, $M_{BA} = 55.38$, $M_{BC} = -55.38$, $M_{CB} = 76.15$, $M_{CD} = -76.15$ kN·m — convergence is rapid (three cycles are enough).
- Check: ΣM at $B = 55.38 - 55.38 = 0$ √; at $C = 76.15 - 76.15 = 0$ √.

Example 4 — Beam with an overhang

Problem

$AB = 6$ m simply supported at A and B with UDL 10 kN/m; overhang $BC = 2$ m with 20 kN at C.

- Overhang moment at B: $M_{BC} = -20 \times 2 = -40$ (known); stiffness of the cantilever = 0, so $DF_{BA} = 1.0$.
- $F'_{BA} = wL^2/8 = +45$. Unbalanced at $B = 45 - 40 = +5 \rightarrow$ balancing $= -5 \rightarrow M_{BA} = +40$, $M_{BC} = -40$ (hogging 40 kN·m).
- $R_A = 30 - 40/6 =$ **23.33 kN**; $R_B = 36.67 + 20 =$ **56.67 kN** √ (same as Unit IV, Example 15).

6. Continuous Beams with Settlement of Supports

- A settlement Δ of a support introduces additional FEMs in the adjacent spans (chord rotation): $FEM = \pm 6EI\Delta/L^2$ for both ends of a span (same sign at both ends).
- **Sign:** when the right end B is **lower** than A, the chord rotates **clockwise** : $FEM_{AB} = FEM_{BA} = -6EI\Delta/L^2$ (anticlockwise on the member ends). When B is **higher** : $+6EI\Delta/L^2$.
- Add the settlement FEM to the FEM due to loads, then distribute as usual.
- If the settlement is given as a **ratio of EI** (e.g., $\Delta = 10$ mm, $EI = 30000$ kN·m²), compute $6EI\Delta/L^2$ numerically first.

Example 5 — Middle support settles

Problem

Two equal spans AB = BC = 6 m, UDL 20 kN/m, EI = 30000 kN·m², A and C hinged; B settles by 10 mm.

- $6EI\Delta/L^2 = 6 \times 30000 \times 0.01 / 36 =$ **50 kN·m** .
- Span AB (B lower): $FEM_{AB} = -60 - 50 = -110$; $FEM_{BA} = +60 - 50 = +10$.
- Span BC (C higher than B): $FEM_{BC} = -60 + 50 = -10$; $FEM_{CB} = +60 + 50 = +110$.
- Modified (A, C hinged): $F'_{BA} = 10 + \frac{1}{2}(110) =$ **+ 65** ; $F'_{BC} = -10 - \frac{1}{2}(110) =$ **− 65** .
- $DF_{BA} = DF_{BC} = 0.5$. Unbalanced moment at B = 65 − 65 = 0 → **$M_{BA} = +65$, $M_{BC} = -65$ kN·m** .
- Compare: without settlement $M_B = 90$ kN·m; **settlement reduces the support moment by 25 kN·m** √ (same as Unit III and Unit IV).
- $R_A = 60 - 65/6 = 49.17$ kN; $R_B = 141.67$ kN.

Example 6 — Settlement of an end support (outline)

- Fixed beam A–B with support B settled by Δ: $FEM_{AB} = FEM_{BA} = -6EI\Delta/L^2$ — since both ends are fixed no distribution takes place; these are the final moments (Unit III).
- For a continuous beam, if only C (end support) sinks, only the span BC gets the extra FEMs; B is then balanced with a non-zero unbalanced moment.

7. Symmetry in Moment Distribution

- For a **symmetrical structure with symmetrical loading** , the rotations at corresponding joints are equal and opposite. Use half the structure with modified stiffness:

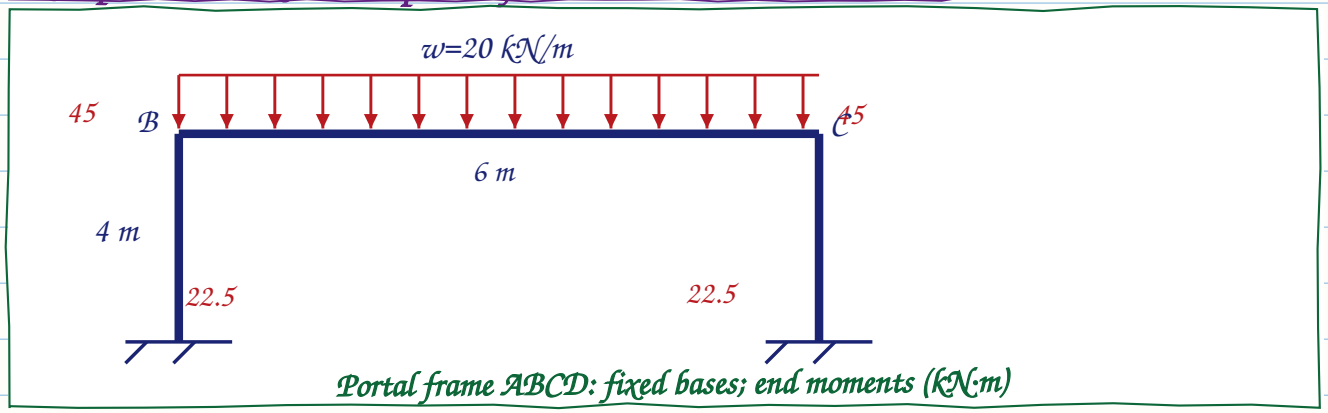
Condition	Stiffness of the symmetric span (far end rotates in opposite direction)	Carry-over
Symmetric loading	2EI/L	none to the far end (use mirror image)
Antisymmetric loading	6EI/L	none to the far end

- This reduces the number of joints to be balanced and often yields the answer in **one cycle** .

8. Single-Bay Portal Frames without Sway

- Frame is **without sway** if (i) the loading and the frame are symmetrical about the vertical axis, or (ii) lateral movement is prevented by a support.
- All members have **no chord rotation** ; fixed end moments act only on loaded members (columns usually have none).
- Steps: DF at B and C, FEM of the beam, balance joints, carry over to the column bases, repeat.

Example 7 — Fixed-base portal frame with UDL on the beam



Portal frame ABCD: fixed bases; end moments (kN·m)

- $h = 4$ m, $L = 6$ m, EI constant. Use symmetry: **beam stiffness $= 2EI/L = 0.333\,EI$; column stiffness (base fixed) $= 4EI/h = EI$.**
- **$DF_{BA} = 1/(1 + 0.333) = 0.75$; $DF_{BC} = 0.25$.**
- $FEM_{BC} = -wL^2/12 = -60$ (and $FEM_{CB} = +60$ by symmetry).

Member end	AB	BA	BC
DF	—	0.75	0.25
FEM	0	0	− 60
Balance at B (+ 60)		+ 45	+ 15
Carry-over (to base A)	+ 22.5		
Final	**+ 22.5**	**+ 45**	**− 45**

- By symmetry: $M_{CB} = +45$, $M_{CD} = -45$, $M_{DC} = -22.5$ kN·m.
- Beam centre $BM = wL^2/8 - 45 =$ **+ 45 kN·m** ; $H_A = H_D = (22.5 + 45)/4 =$ **16.88 kN**; $V = 60$ kN.
- √ Same as the slope-deflection result (Unit IV, Example 7).

Example 8 — Hinged-base portal frame (same loading)

- Column stiffness (base hinged) $= 3EI/4 = 0.75\,EI$; beam (symmetric) $= 0.333\,EI$. **$DF_{BA} = 0.75/1.083 = 0.692$; $DF_{BC} = 0.308$.**
- Balance at B: + 60 → BA = + 41.54, BC = + 18.46 → **$M_{BC} = -60 + 18.46 = -41.54$; $M_{BA} = +41.54$; $M_{AB} = 0$** (no carry-over to a hinge).
- Beam centre $BM = 90 - 41.54 =$ **+ 48.46 kN·m** ; $H = 41.54/4 =$ **10.38 kN**.

Example 9 — Fixed-base frame, central point load P = 48 kN on the beam

- Same DF (0.75, 0.25). $FEM_{BC} = -PL/8 = -36$. Balance: + 36 → BA = + 27, BC = + 9.
- **$M_{BA} = 27$, $M_{AB} = 13.5$, $M_{BC} = -27$ kN·m** ; beam centre $= PL/4 - 27 =$ **+ 45 kN·m** ; $H =$ 10.13 kN.

Example 10 — Beam stiffer than columns ($I_{beam} = 2\,I_{col}$)

- $K_{col} = 4EI/4 = EI$; $K_{beam} = 2\,(2EI)/6 = 0.667\,EI$ → **$DF_{BA} = 0.6$; $DF_{BC} = 0.4$** .
- Balance of + 60: BA = + 36, BC = + 24 → **$M_{BA} = 36$, $M_{AB} = 18$, $M_{BC} = -36$ kN·m** (compare

Unit IV, Example 8).

Note for frames with sway

- If the frame sways (unsymmetrical loading, lateral loads), first analyse the **non-sway** frame (with an imaginary support), then give the sway displacement and correct the moments using the **shear equation**; this is beyond this unit's requirement.

8A. More Solved Examples — Beams

Example 11 — Both ends fixed, two spans

Problem

ABC; A and C fixed; AB = 6 m (UDL 20 kN/m); BC = 4 m (central load 40 kN); EI constant.

- $K_{BA} = 4EI/6 = 0.667\,EI$; $K_{BC} = 4EI/4 = EI$ → **$DF_{BA} = 0.4$, $DF_{BC} = 0.6$**.
- FEM: AB = − 60, BA = + 60, BC = − 20, CB = + 20.

Member end	AB	BA	BC	CB
DF	—	0.4	0.6	—
FEM	− 60	+ 60	− 20	+ 20
Balance B (unbalanced + 40)		− 16	− 24	
Carry-over	− 8			− 12
Final	**− 68**	**+ 44**	**− 44**	**+ 8**

- Reactions: span AB: $R_A = 60 + (68 - 44)/6 =$ **64 kN**, R_B(left) = 56 kN; span BC: R_B(right) = 20 + (44 − 8)/4 = 29 kN, $R_C = 20 - 9 =$ **11 kN**. **$R_B = 85$ kN**; check 64 + 85 + 11 = 160 √.
- √ matches slope-deflection: $EI\theta_B = -24$.

Example 12 — Different EI in the two spans

Problem

Same loading, but BC has moment of inertia 2I (AB has I); A and C fixed.

- $K_{BA} = 4EI/6 = 0.667\,EI$; $K_{BC} = 4(2EI)/4 = 2\,EI$ → **$DF_{BA} = 0.25$, $DF_{BC} = 0.75$** (stiffer span attracts more moment).
- Unbalanced + 40 → balance: BA = − 10, BC = − 30; carry-over: AB = − 5, CB = − 15.

Final moments

$M_{AB} = -65$, $M_{BA} = +50$, $M_{BC} = -50$, $M_{CB} = +5$ kN·m

Example 13 — Propped cantilever with a central point load

- A fixed, B roller: the far end is hinged so use modified FEM at A: $F'_{AB} = -PL/8 - \tfrac{1}{2}(PL/8) =$ **$-3PL/16$**. With one joint and a fixed end, $M_A = -3PL/16$ (hogging) directly; $R_B = 5P/16$ √.

Reactions of Example 3 (three-span beam)

Span	Left reaction	Right reaction
AB (M_A = 62.31, M_B = 55.38)	60 + (62.31 − 55.38)/6 = 61.16	58.84
BC (M_B = 55.38, M_C = 76.15)	60 + (55.38 − 76.15)/6 = 56.54	63.46
CD (M_C = 76.15, M_D = 0)	60 + 76.15/6 = 72.69	47.31

- **R_A = 61.16, R_B = 115.38, R_C = 136.15, R_D = 47.31 kN** (sum = 360 = 3 × 120 ✓).

8B. Distribution Factors — Illustration

- Joint B of a frame: column BA (I, h = 4 m, base fixed), beam BC on the left (L = 6 m), beam BD on the right (L = 5 m), all the same E and I; far ends fixed.

Member	Stiffness $4EI/L$	Relative K	DF
BA (column)	$4EI/4$	1.000	0.405
BC (beam 6 m)	$4EI/6$	0.667	0.270
BD (beam 5 m)	$4EI/5$	0.800	0.324
Total		2.467	1.000

- If the far end of BD were hinged: $K_{BD} = 3EI/5 = 0.6$ (DF changes to 0.441, 0.294, 0.265 → sum = 1).
- **Stiffer members take the larger share** of the unbalanced moment.

8C. Kani's Method (brief)

- Kani (1947) — a **rotation-contribution method**; self-correcting iterative method for frames and beams.
- Rotation contribution of joint i on member ij: $M'_{ij} = \mu_{ij} (\Sigma M^F_i + \Sigma M'_{ni})$, with rotation factor $\mu_{ij} = -\tfrac{1}{2} K_{ij} / \Sigma K_i$; final end moment $M_{ij} = M^F_{ij} + 2M'_{ij} + M'_{ji}$.
- Same answers as moment distribution; used mostly when many storeys are involved.

8D. Full Iteration for the Portal Frame (without using symmetry)

- The same frame as Example 7 (h = 4 m, L = 6 m, fixed bases, UDL 20 kN/m) solved by the plain method — columns $K = 4EI/4 = EI$, beam $K = 4EI/6 = 0.667\,EI$ → **DF_{BA} = 0.6, DF_{BC} = 0.4; DF_{CB} = 0.4, DF_{CD} = 0.6**.

Step	AB	BA	BC	CB	CD	DC
DF	—	0.6	0.4	0.4	0.6	—

Step	AB	BA	BC	CB	CD	DC
FEM	0.00	0.00	-60.00	+60.00	0.00	0.00
Bal B(1)		+36.00	+24.00			
CO(1b)	+18.00			+12.00		
Bal C(1)				-28.80	-43.20	
CO(1c)			-14.40			-21.60
Bal B(2)		+8.64	+5.76			
CO(2b)	+4.32			+2.88		
Bal C(2)				-1.15	-1.73	
CO(2c)			-0.58			-0.86
Bal B(3)		+0.35	+0.23			
CO(3b)	+0.17			+0.12		
Bal C(3)				-0.05	-0.07	
CO(3c)			-0.02			-0.03
Final	+22.50	+45.00	-45.00	+45.00	-45.00	-22.50

- The cycles converge to **22.50, 45.00, – 45.00, + 45.00, – 45.00, – 22.50** — identical to the symmetric shortcut (Example 7) and to the slope-deflection result.
- Lesson: using **symmetry** replaces about 6 balancing steps by a single one.

Check of the final moments

- Joint B: 45 – 45 = 0 √. Joint C: 45 – 45 = 0 √.
- Horizontal reaction: $H = (22.5 + 45)/4 = 16.88$ kN (equal and opposite at A and D √).
- Vertical reaction: $V = 60$ kN each ($\Sigma V = 120 = 20 \times 6$ √).
- Beam BC: centre BM = 90 – 45 = **+ 45 kN·m** ; points of contraflexure from $M(x) = 60x – 10x^2 – 45 = 0 \rightarrow x = 3 \pm 2.12 \rightarrow$ **0.88 m and 5.12 m** from B.

9. Reactions, Shear Force and Bending Moment Diagrams

- For every span: $R_{left} = R_{free} + (M_{leftend} – M_{rightend})/L$ (with hogging end moments positive).
- Draw SFD from the reactions; BM at any section = free BM – (linear variation of end moments).
- **Point of contraflexure:** where BM changes sign; **maximum sagging** where SF = 0.
- In frames, draw BMD on the **tension side** ; verify joint equilibrium and horizontal-force equilibrium at the bases.

10. Accuracy and Termination

- Continue cycles until the **carry-over moments are less than about 1%** of the FEM; the error in final moments is usually within 1–2%.
- Convergence is fast when the structure is not highly redundant; each carry-over halves the quantity.
- **Check:** (i) $\Sigma DF = 1$ at each joint; (ii) ΣM at each joint = 0; (iii) ΣV = total load.

11. Comparison — Slope-Deflection vs Moment Distribution

Basis	Slope-deflection	Moment distribution
Nature	exact; equations	approximate (iterative) but converges to exact
Unknowns	joint rotations θ (and sway)	none solved directly
Equations	simultaneous equations	no equations
Effort	heavy for many joints	easier for many joints
Settlement	included through ψ	included through extra FEM
Checks	substitute back	table totals, joint equilibrium
Developed by	G. A. Maney (1915)	Hardy Cross (1930)

12. Advantages, Limitations and Related Methods

- **Advantages:** simple tabular work; no simultaneous equations; errors self-correct; physical meaning is clear; good for hand design of continuous beams.
- **Limitations:** convergence slows for large frames; sway frames need extra steps; not convenient for computer programming compared to matrix methods.
- **Kani's method:** a similar rotation-contribution iteration that is self-correcting.
- **Matrix stiffness method:** the computer-based extension (K = member stiffness $4EI/L$, $3EI/L$, $6EI/L^2$, $12EI/L^3$).

13. Common Mistakes and Exam Tips

No.	Point
1	ΣDF at every joint = 1; a fixed support has $DF = 0$.
2	Use $3EI/L$ (and modified FEM) for a hinged far end and do **not** carry over.
3	Balancing moment = − (unbalanced moment) × DF — sign change!
4	Carry over = ½ × balancing moment, to the **far end of the same member**.
5	Add settlement FEM = $6EI\Delta/L^2$ with the correct sign (B lower → anticlockwise).
6	Do not balance a fixed end; do not forget the known overhang moment.

No.	Point
7	Draw a neat table with joints, DFs, FEMs, cycles and a final sum row.
8	Verify by reactions: ΣV = total load.

14. Practice Problems with Answers

No.	Problem	Answer
1	Two equal spans 5 m, UDL 24 kN/m, ends hinged	M_B = 75 kN·m
2	Two equal spans 6 m, central loads 60 kN each, ends hinged	$M_B = 3PL/16$ = 67.5 kN·m
3	Fixed beam 6 m, UDL 20 kN/m, MDM check	$M_A = M_B$ = 60 kN·m
4	Portal frame h = 4 m, L = 6 m, fixed bases, UDL 30 kN/m on the beam	M_{BA} = 67.5; M_{AB} = 33.75 kN·m
5	Two equal spans 6 m, w = 20, B settles 10 mm, EI = 30000	M_B = 65 kN·m
6	Three equal spans 6 m, UDL 10 kN/m, ends hinged	$M_B = M_C = wL^2/10$ = 36 kN·m

15. Quick Revision — Unit V

Key formulae

$K = 4EI/L$ (far end fixed), $3EI/L$ (hinged) | $COF = ½$ | $DF = K/\Sigma K$, $\Sigma DF = 1$ | Balancing moment = − unbalanced × DF | Carry-over = ½ × balancing moment | Modified FEM (hinged far end): $F - ½F_{far}$ | Settlement FEM = $\pm 6EI\Delta/L^2$ | Symmetric span: $K = 2EI/L$.

Possible exam questions

1. Explain the moment distribution method; define stiffness, carry-over factor and distribution factor.
2. Analyse a continuous beam by the moment distribution method and draw SFD and BMD.
3. Analyse a continuous beam with settlement of supports by MDM.
4. Analyse a single-bay portal frame without sway by MDM; draw the BMD.
5. Compare slope-deflection and moment distribution methods.